



University of Moratuwa

Department of Electronic and Telecommunication Engineering

Final Report

Integrated Precision Cinnamon Processor (IPCP)

A Smart Electronic Cinnamon Grading System

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1. Problem Description, Motivation and Justification

1.1 Background

Sri Lanka supplies 85–90% of the world’s true cinnamon (*Cinnamomum verum*), yet a significant share of the value that this position could generate is lost due to inconsistent quality grading at the point of primary processing. The current grading practice relies almost entirely on manual visual inspection by trained workers who sort quills by eye and hand-feel. This method is inherently subjective: two graders can assign different grades to the same quill, and fatigue, lighting conditions, and experience level all introduce further variance.

No affordable electronic grading instrument currently exists that has been purpose-built and calibrated for cinnamon quills. As a direct consequence, an estimated 90% of Sri Lanka’s cinnamon is exported in raw, ungraded quill form, forfeiting the price premium that objectively graded product could command.

1.2 Why This Problem Matters

Grade is not a cosmetic distinction – it is directly tied to farm-gate and export price. Under the Sri Lanka Standards Institute’s SLS 81:2021 specification, **Alba** grade (the highest, defined by a nominal diameter of approximately 6 mm measured against a standard pencil gauge) commands roughly three times the price of the lower C/M/H grades. A farmer or small processor who cannot reliably and repeatably determine which grade a given quill belongs to has no way to capture this premium, and buyers discount uncertain lots as a matter of course.

1.3 Justification for Selecting This Problem

- **Scale of impact:** Sri Lanka’s dominant position in the global true-cinnamon trade means that even a modest improvement in grading accuracy at the smallholder/SME level translates into a large aggregate gain in export value.
- **Clear, quantifiable standard to design against:** SLS 81:2021 defines grading purely in terms of physical, machine-measurable parameters – diameter (via sticks-per-kg and pencil-gauge comparison) and moisture content ($\leq 14\%$ for quills) – which makes the problem tractable for a low-cost electromechanical instrument rather than requiring subjective sensory judgement.
- **Gap in the market:** Existing quality-control technology for spices is either manual (visual inspection) or built around generic optical/imaging platforms not adapted to the mechanical properties of cinnamon bark (see Section 4 for why an optical approach was evaluated and rejected).
- **Accessibility of a solution:** The parameters that define the grade (mass and diameter) can be captured with low-cost, readily available sensors (load cells, a stepper driven caliper mechanism), keeping the bill-of-materials within reach of an SME processing shed rather than requiring laboratory-grade equipment.



(a) .



(b)



(c)



(d)

Figure 1.1
Traditional Method

2. Technical Feasibility

2.1 Resource Requirements

All core components used in the IPCP are commercially available, low-cost, off-the-shelf modules, which keeps both the engineering risk and the bill-of-materials cost low:

- ESP32 (dual-core) microcontroller – widely available, well documented, sufficient processing headroom for simultaneous interrupt handling and motor control.
- NEMA 17 stepper motor with a TMC2208 (StealthChop) driver – a mature, silent-running combination commonly used in 3D-printing and CNC applications, meaning tooling, drivers, and community support are mature.
- Dual HX711 24-bit ADC modules paired with load cells – inexpensive, high-resolution sensing components already proven in low-cost weighing applications.
- Standard OLED display, buck converter, and a 12 V DC bench-style power supply.

Total estimated bill-of-materials cost for a single prototype unit is approximately **Rs. 21,000** (see Chapter 9 for the full breakdown), which is realistic for a working prototype at student-project scale and leaves margin for a commercially viable selling price.

2.2 Performance Targets and Reachability

The system’s core performance requirement is to resolve quill diameter to the tolerances implied by SLS 81:2021’s grade boundaries (differences on the order of a few millimetres between adjacent grades). Because diameter is derived from stepper motor step count via the relation

$$D = 50 - L$$

(where L is the linear travel of the jaw in millimetres over a 50 mm measurement range), the achievable resolution is set by the stepper’s microstepping configuration rather than by any noisy analog measurement – this is a fundamentally repeatable, digital measurement, which makes the diameter-grading target reachable with standard components. Mass measurement resolution is similarly bounded by the load cell and HX711 combination, which is well within the tolerance needed to distinguish grade bands.

2.3 Repeatability Testing

To validate that the performance targets described above are actually reachable in practice, repeatability trials were run by taking 10 successive measurements of the same physical quill sample for three different grades (C4, C3, and M4), recording both diameter and mass on each pass.

Across all three grades, both diameter and mass readings stay tightly clustered around a consistent mean over the 10 repetitions, with only small run-to-run variation and no drift

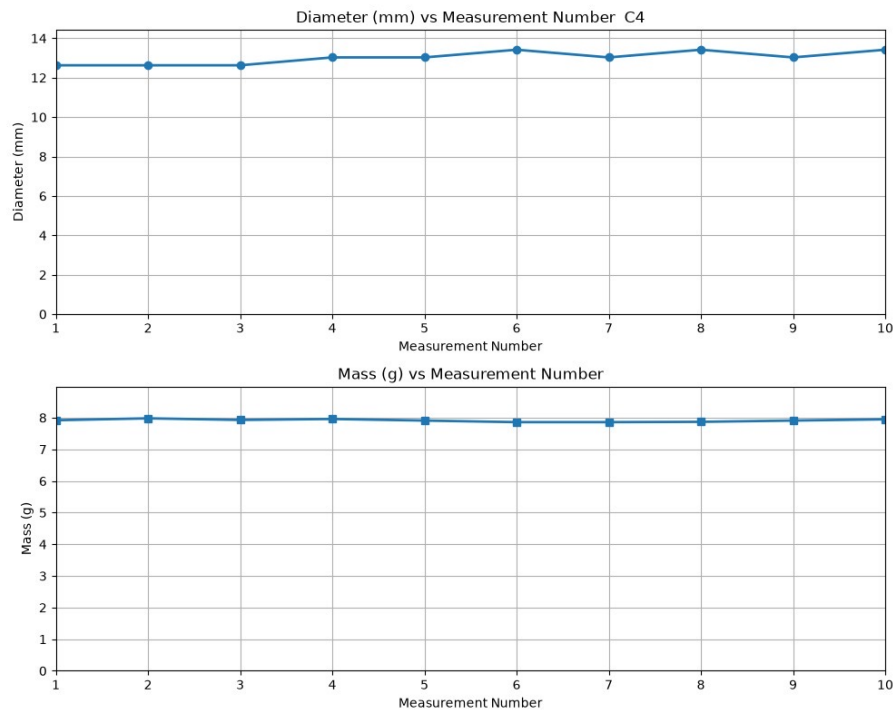


Figure 2.1: C4 grade: diameter and mass across 10 repeated measurements.

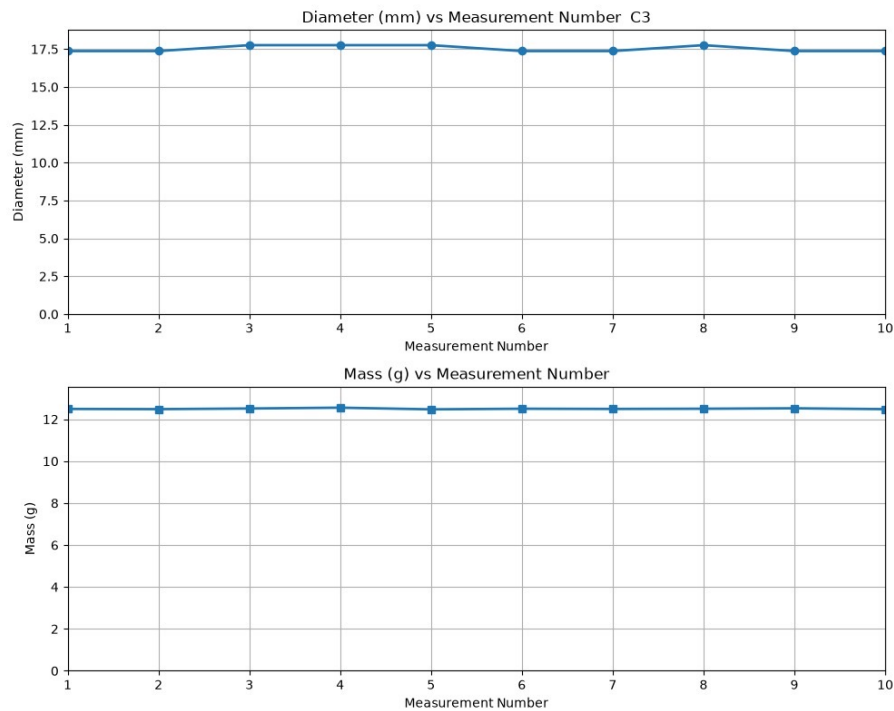


Figure 2.2: C3 grade: diameter and mass across 10 repeated measurements.

over the measurement sequence. This is direct empirical evidence that the mechanical, contact-triggered measurement approach (Section 4) produces stable, repeatable readings for a given quill – confirming that the diameter-based grade thresholds defined in SLS 81:2021 can reliably be resolved by this measurement mechanism, rather than this being only a theoretical expectation.

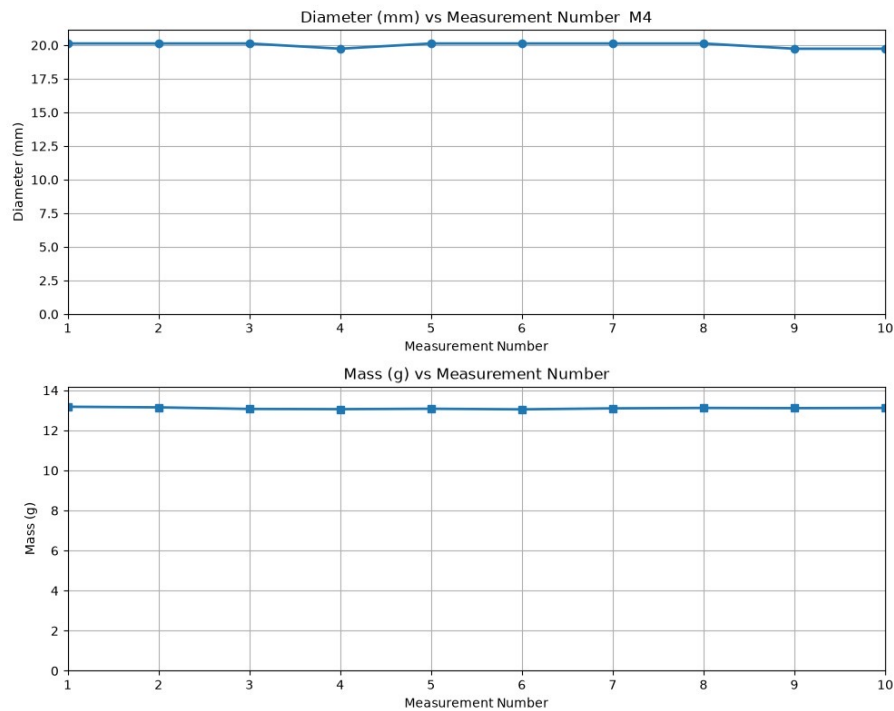


Figure 2.3: M4 grade: diameter and mass across 10 repeated measurements.

2.4 Deviations from the Original Plan

The original concept explored a non-contact optical diameter-sensing approach before settling on the mechanical, load-cell-based design that was ultimately built. This change of direction, and the reasoning behind it, is detailed in Section 4. The key lesson driving the deviation was that a method which performs well in a clean lab environment (optical shadow profiling) was not robust enough for the dusty, high-organic-debris environment of a cinnamon processing shed – reinforcing a design philosophy of favouring simple, mechanically robust sensing over more elegant but fragile alternatives.

3. Product Architecture

3.1 System Overview

The IPCP is a feedback-driven electromechanical system governed by an ESP32 microcontroller, designed to convert physical displacement and mass into a standardised DCD (Diameter/Class/Density) grade per SLS 81:2021. The operating flow is: a farmer places a cinnamon quill into the measurement channel; the system measures weight and diameter; the ESP32 processes both readings against pre-programmed grade thresholds; and the resulting grade, weight, and diameter are displayed instantly on an onboard OLED screen.

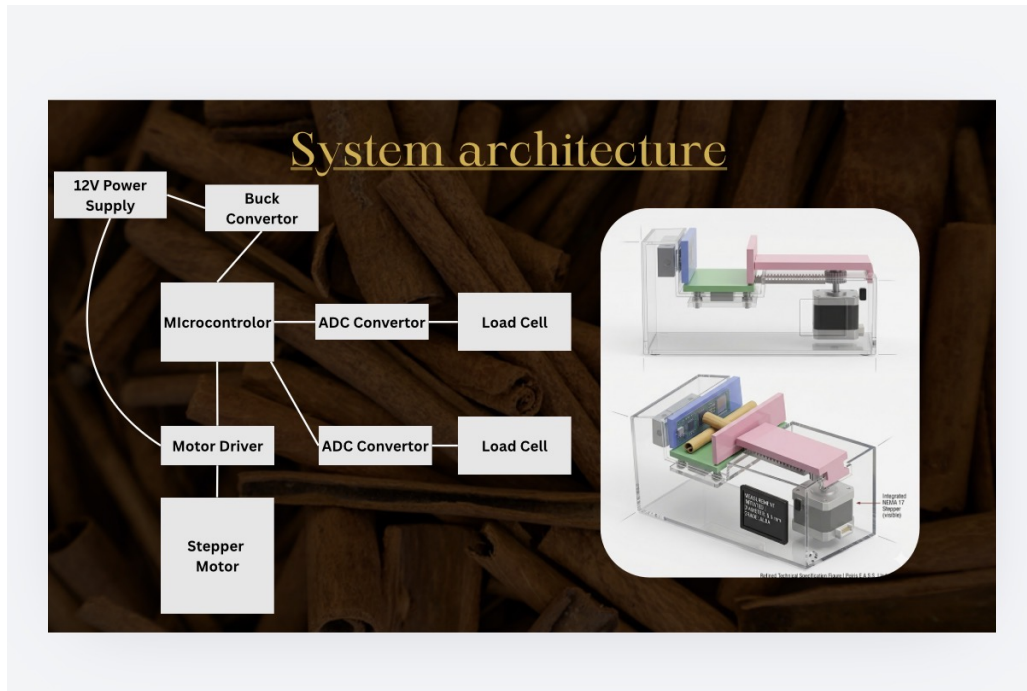


Figure 3.1: IPCP block-level system architecture.

3.2 Block Descriptions

12V Power Supply & Buck Converter

Supplies the raw 12 V rail used by the stepper motor driver, stepped down through a buck converter to the regulated logic-level voltage required by the microcontroller and sensor modules.

Microcontroller (ESP32, dual-core)

The central controller. Handles high-speed processing of sensor interrupts and simultaneous motor control, and is IoT-ready for future data logging.

Motor Driver (TMC2208, StealthChop)

Drives the NEMA 17 stepper with silent, micro-stepped motion, chosen specifically to eliminate mechanical noise that would otherwise corrupt the sensitive load-cell readings.

Stepper Motor (NEMA 17)

Provides high holding torque and precise step resolution, driving a linear rack-and-pinion assembly that moves a jaw across a 50 mm measurement range.

ADC Converters (HX711, 24-bit) + Load Cells (x2)

One load cell forms the measurement floor for mass recording (weighing); the second is mounted on the stationary wall of the measurement channel as a compression load cell, used purely for haptic contact detection (see diameter measurement principle below) rather than for weight.

3.3 Diameter Measurement Principle

Figure 3.2: Measurement channel: resting state and calibration (system zero) state.

As the jaw advances toward a single cinnamon quill placed in the channel, the compression load cell detects the moment physical contact is made and triggers a high-frequency interrupt via the 24-bit HX711 ADC. This immediately halts the TMC2208 driver at the precise point of contact. Diameter is then computed from the recorded stepper motor step count using $D = 50 - L$, and the resulting value, together with the independently measured mass, is compared against the pre-programmed SLS 81:2021 grade thresholds before being pushed to the OLED display.

4. Alternatives Considered

4.1 The Optical Method

An earlier design direction considered a non-contact optical technique, in which a high-resolution diffused LED light source would project a quill's shadow onto a linear CCD sensor, with diameter calculated by counting occluded pixels (shadow width).



Figure 4.1: Optical shadow-profiling concept, rejected in favour of the mechanical approach.

4.1.1 Critical Rejection Reasons

- **Dust interference:** Cinnamon processing sheds have high levels of organic dust, which settles on lenses and creates false shadows, rendering the sensor inaccurate in real operating conditions.
- **Bark profile bias:** Optical systems measure surface “fuzz” (loose bark fibre) rather than the structural diameter that the grading standard actually cares about, leading to inconsistent DCD grading outcomes.
- **Non-integrated sensing:** Unlike the mechanical system that was ultimately built, a CCD-based approach cannot weigh the quill, requiring a redundant second measurement step and therefore added cost and complexity for no accuracy benefit.

The mechanical, load-cell-based design was selected instead because it is materially more robust to the field environment (dust, vibration, humidity) while measuring both diameter and mass with a single integrated mechanism.

5. PCB Design

5.1 Schematic Design

The schematic was developed in Altium Designer and is organised into clearly separated functional blocks: buttons/firmware-upload support, OLED display, dual HX711 load-cell interfaces, the ESP32 core with USB, the power section (12 V input through the buck converter), and the TMC2208 motor driver stage.

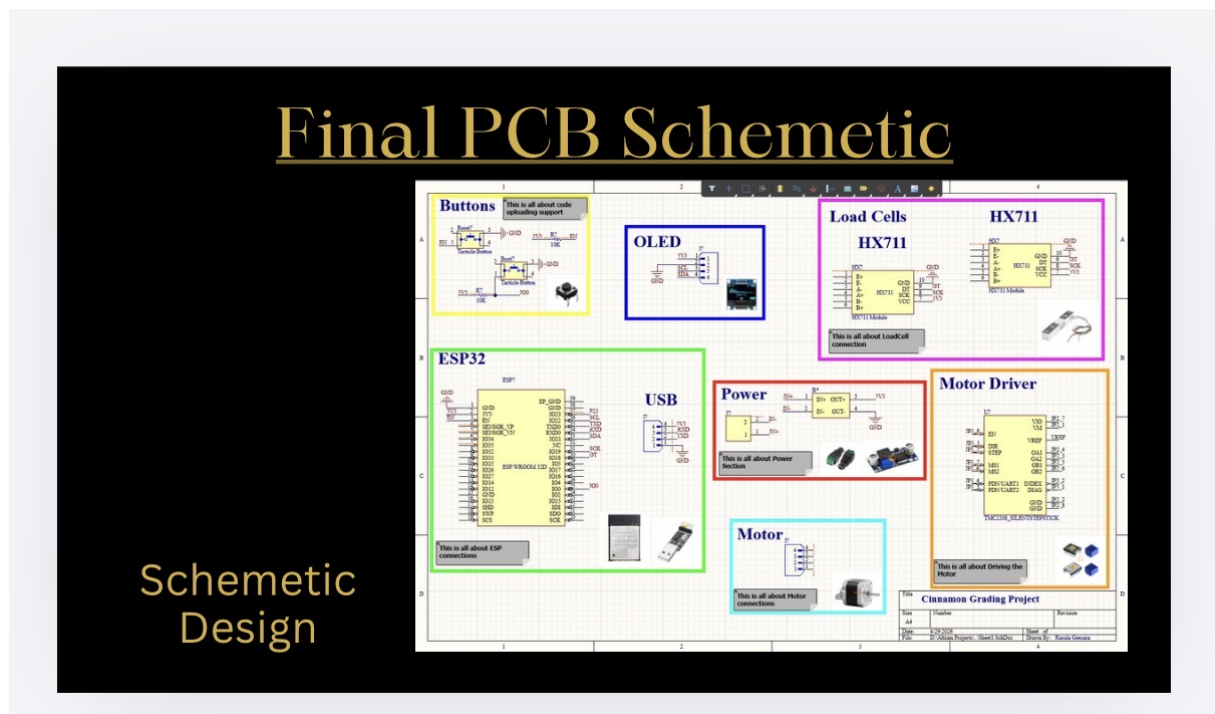


Figure 5.1: Final PCB schematic, organised by functional block.

5.2 Layout Design

The board was laid out as a **2-layer PCB**. The layout groups components into the same functional zones used in the schematic: the two HX711 load-cell interface circuits are placed side by side on the left, the ESP32-WROOM-32 module occupies the top-right with its connector headers broken out along the board edge, and the buck-converter power stage (12 V input, regulated output) sits along the bottom-left, physically separated from the digital/analog signal circuitry above it.

5.3 Justification for the PCB Stack

- **2-layer stack-up:** a 2-layer board (top copper for signal routing, bottom copper used substantially as a ground plane/return path) was chosen over a single-sided board to

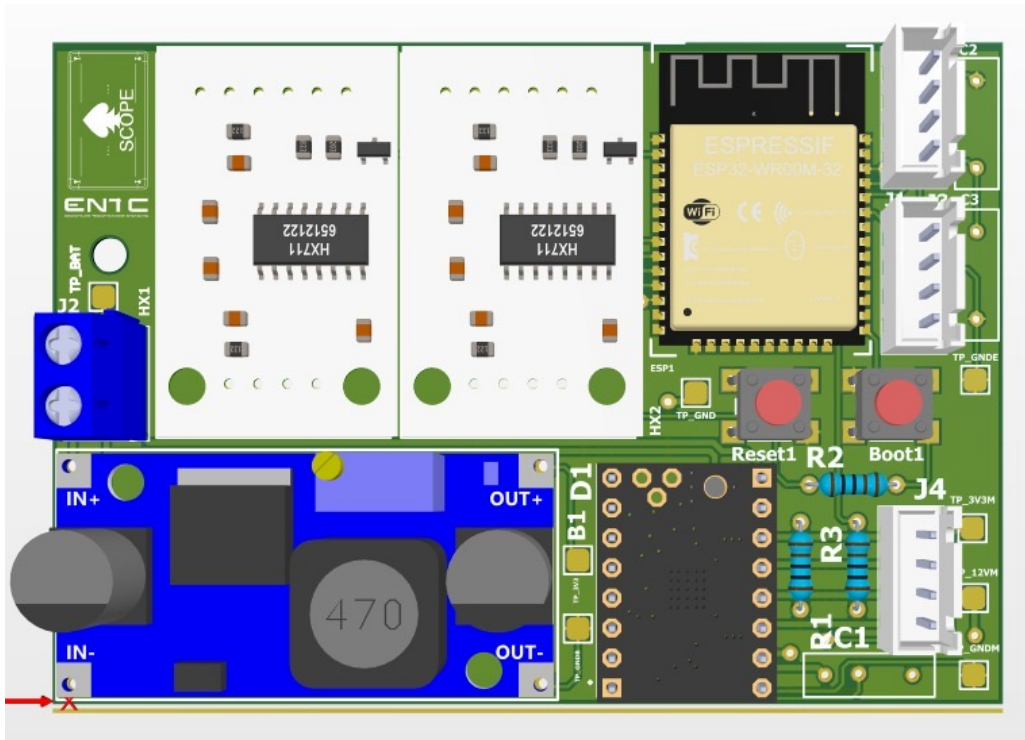


Figure 5.2: Final 2-layer PCB layout render, showing the dual HX711 load-cell interfaces (left), ESP32-WROOM-32 module (top right), and buck-converter power stage (bottom left).

give the HX711 analog front-end a clean, continuous ground reference – important for resolving small load-cell signal changes – while keeping fabrication cost low compared to a 4-layer design, which was not necessary at this signal complexity and prototype volume.

- Analog front-end (HX711 + load cells) was kept as a distinct, clearly bounded block in both the schematic and the physical layout, to simplify isolating it from digital switching noise generated by the stepper driver.
- The TMC2208 was selected specifically for its StealthChop silent-drive mode, which reduces electrical and mechanical noise coupling into the sensitive load-cell signal path – a decision that directly shaped the PCB partitioning (motor driver connector placed away from the load-cell block).
- The buck-converter power stage was placed at the board edge, physically separated from the HX711/ESP32 circuitry, isolating the noisier 12 V motor rail and switching regulator from the regulated logic supply and analog front-end.

6. Enclosure Design

6.1 Final Enclosure



Figure 6.1: Final 3D-printed enclosure with a cinnamon quill loaded in the measurement channel.

The final enclosure exposes the quill loading channel, houses the OLED readout on the top face, and integrates the NEMA 17 stepper and rack mechanism within a compact two-part 3D-printed housing.

7. Final Product Details

7.1 Specifications

Parameter	Specification
Microcontroller	ESP32 (dual-core)
Motion actuator	NEMA 17 stepper motor
Motor driver	TMC2208 (StealthChop, silent micro-stepping)
Dimension sensing	HX711 + compression load cell (haptic contact detection)
Weight sensing	Independent platform load cell + HX711
Measurement range	50 mm linear jaw travel
Display	OLED
Power input	12 V DC supply, stepped down via buck converter
Grading reference	SLS 81:2021 (Alba, C, M, H grade bands)

Table 7.1: IPCP final specifications.

8. Manufacturing Considerations for Volume Production

Moving from a single prototype to volume production requires attention to the following factors:

- **Design for Manufacture (DFM):** the 3D-printed enclosure used for the prototype would need to be redesigned for injection moulding (draft angles, wall thickness, gate/rib placement) to be cost-effective at volume.
- **Component sourcing and lead times:** HX711 modules, load cells, and TMC2208 drivers are currently sourced as hobbyist/breakout modules; volume production would require sourcing bare ICs and qualifying alternate/second-source suppliers to avoid single-source risk.
- **Calibration workflow:** every unit requires per-unit calibration of the load cells and the mechanical zero-point of the jaw; a repeatable factory calibration jig and procedure is required rather than the manual calibration used on the prototype.
- **PCB assembly:** moving from a hand/lab-assembled board to a fabricated, reflow-soldered PCB assembly (PCBA) with automated optical inspection (AOI) for quality control.
- **Environmental robustness:** the target environment (processing sheds with organic dust, humidity, and variable temperature) motivates an ingress-protected enclosure rating and dust-sealing around the mechanical jaw opening.
- **Regulatory/compliance:** electrical safety and EMC compliance for a mains- or DC-adaptor-powered device intended for commercial sale.
- **After-sales support:** given the maintenance-contract element of the revenue model (Chapter 10), a serviceable design (accessible load cells, replaceable driver board) reduces field-service cost.

9. Costing

9.1 Prototype Bill-of-Materials Cost

The prototype was fabricated using individually purchased components, 3D-printed mechanical parts, and manually assembled PCBs. Therefore, the manufacturing cost is relatively high compared to commercial production.

Component Group	Prototype Cost (Rs.)
NEMA 17 Stepper + TMC2208 Driver	4,000
ESP32 Controller + OLED Interface	3,500
Dual Load Cells + HX711 Modules	2,000
Mechanical Frame & Fabrication	8,000
PCB, Power Supply & Assembly	4,500
Total Manufacturing Cost	21,000

Table 9.1: Prototype manufacturing cost.

9.2 Estimated Cost for Volume Production

For commercial manufacturing, production is assumed at approximately 1000 units per batch. Bulk purchasing of electronic components, automated PCB assembly, and injection-moulded enclosures significantly reduce the manufacturing cost.

Component Group	Prototype (Rs.)	Bulk Production (Rs.)
NEMA 17 Stepper + TMC2208 Driver	4,000	3,200
ESP32 Controller + OLED Interface	3,500	2,600
Dual Load Cells + HX711 Modules	2,000	1,500
Mechanical Frame & Enclosure	8,000	4,500
PCB, Power Supply & Assembly	4,500	3,000
Manufacturing Cost	21,000	14,800

Table 9.2: Comparison of prototype and volume-production manufacturing costs.

The manufacturing cost decreases by approximately 30% due to:

- Bulk purchasing of electronic components directly from suppliers.
- Surface-mount automated PCB assembly (SMT).
- Injection-moulded enclosure replacing the 3D-printed enclosure.
- Reduced assembly time and labour costs.
- Standardized factory calibration procedures.

9.3 Recommended Selling Price

Commercial products must recover manufacturing costs while covering quality assurance, calibration, packaging, transportation, warranty, distribution, and company operating expenses.

Based on the estimated manufacturing cost of **Rs. 14,800** per unit, the recommended selling price is

Rs. 29,900

This provides

$$\text{Gross Profit} = 29,900 - 14,800 = 15,100 \text{ Rs.}$$

which corresponds to a gross profit margin of approximately

$$\frac{15,100}{29,900} \times 100 = 50.5\%$$

This pricing keeps the product affordable for cinnamon processors while ensuring commercial sustainability.

9.4 Return on Investment (ROI)

The IPCP enables accurate identification of premium-grade cinnamon, allowing processors to obtain significantly higher market prices. Considering the price difference between premium and lower grades, the investment can typically be recovered after processing approximately 10–15 kg of premium cinnamon, providing an attractive payback period for commercial users..

10. Marketing, Sales and Beyond

10.1 Value Proposition

Objective DCD grading removes human bias from the grading process. The integrated mass and diameter sensing mechanism also reduces labour time by an estimated 40% per quill compared to manual sorting.

10.2 Target Segments

The go-to-market focus is B2B: Southern Province SMEs, regional export processing centres, and large-scale cinnamon cooperatives – i.e. the actors who currently absorb the cost of inconsistent grading and stand to gain the most from an objective, repeatable measurement tool.

10.3 Revenue Model

Direct B2B unit sales, coupled with annual maintenance contracts covering sensor recalibration – an important recurring-revenue stream given the harsh, dusty shed environments the units operate in, which will require periodic load-cell recalibration.

11. Project Budget with Bill of Quantities (BOQ)

11.1 Prototype Manufacturing Cost

Component	Quantity	Cost (Rs.)
NEMA 17 Stepper + TMC2208 Driver	1	4,000
ESP32 Controller + OLED Interface	1	3,500
Dual Load Cells + HX711 Modules	2 Sets	2,000
Mechanical Frame & Fabrication	1	8,000
PCB, Power Supply & Assembly	1	4,500
Total Prototype Cost		21,000

11.2 Estimated Manufacturing Cost for Volume Production (1000 Units)

Component	Cost per Unit (Rs.)	Total for 1000 Units (Rs.)
NEMA 17 Stepper + TMC2208 Driver	3,200	3,200,000
ESP32 Controller + OLED Interface	2,600	2,600,000
Dual Load Cells + HX711 Modules	1,500	1,500,000
Mechanical Frame & Enclosure	4,500	4,500,000
PCB, Power Supply & Assembly	3,000	3,000,000
Manufacturing Cost	14,800	14,800,000

For a production batch of 1000 units, the total manufacturing cost is approximately **Rs. 14.8 million**. At a selling price of **Rs. 29,900** per unit, the total revenue becomes **Rs. 29.9 million**, resulting in an estimated gross profit of approximately **Rs. 15.1 million** before operating expenses, marketing, taxes, and distribution costs.

12. Future Improvements

- **Moisture sensing:** SLS 81:2021 specifies a moisture content threshold of $\leq 14\%$ for quills, but the current prototype grades on diameter and mass only. Adding a moisture sensor would allow the IPCP to verify full compliance with the standard rather than a subset of it.
- **IoT data logging:** the ESP32 is already IoT-ready; connecting each unit to a cloud dashboard would let cooperatives and exporters track grading throughput and quality trends across many units/locations.
- **Automated feeding/sorting:** extending the mechanism to automatically sort graded quills into separate bins would further reduce manual labour time beyond the current 40% reduction.
- **Battery-powered field variant:** for use away from a fixed 12 V supply, e.g. at smaller outgrower collection points.
- **Enclosure ruggedisation:** improved dust/humidity sealing for long-term deployment in shed environments.

13. Task Allocation Among Group Members

Member	Contribution
Maldeniya P.A.D.G.R	Enclosure Design, Business Planning, Assembly, Debugging
Ranasara Maldeniya	Testing, Assembly, Coding, Documentation, Debugging
Subodha Peiris	Testing, Assembly, Schematic Design, PCB Soldering, Coding, Debugging
Dulen Sandiw	Assembly, Enclosure Design, Debugging
Rasula Geesara	PCB Design, PCB Assembly, PCB Soldering, Debugging